Bug Reports — HW2: Debounce / GPIO

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Defects found while testing button debounce on an ESP32 (BUTTON and NOISE inputs on GPIO18, LED on GPIO2). TC-01…04 passed; TC-05 and TC-06 failed.

Environment: ESP32 DevKit v1, Wokwi simulation

Summary

ID	Title	Severity
Bug_1 (TC-05)	Extra logical event/action is present after pressing the NOISE button	Critical
Bug_2 (TC-06)	Extra logical event/action is present after pressing the NOISE button during a critical operation	Critical

Bug_1 (TC-05) — Extra logical event/action is present after pressing the NOISE button

Environment	ESP32 DevKit v1
Preconditions	Simulation is started
Steps to Reproduce	1. Press the NOISE button. 2. Release the NOISE button. 3. Check the Serial Monitor.
Expected	No false logical press/action
Actual	A logical press/action is present after pressing the NOISE button
Evidence	TC-05 Short noise pulse (screenshot below)
Reproducibility	Always
Severity	Critical

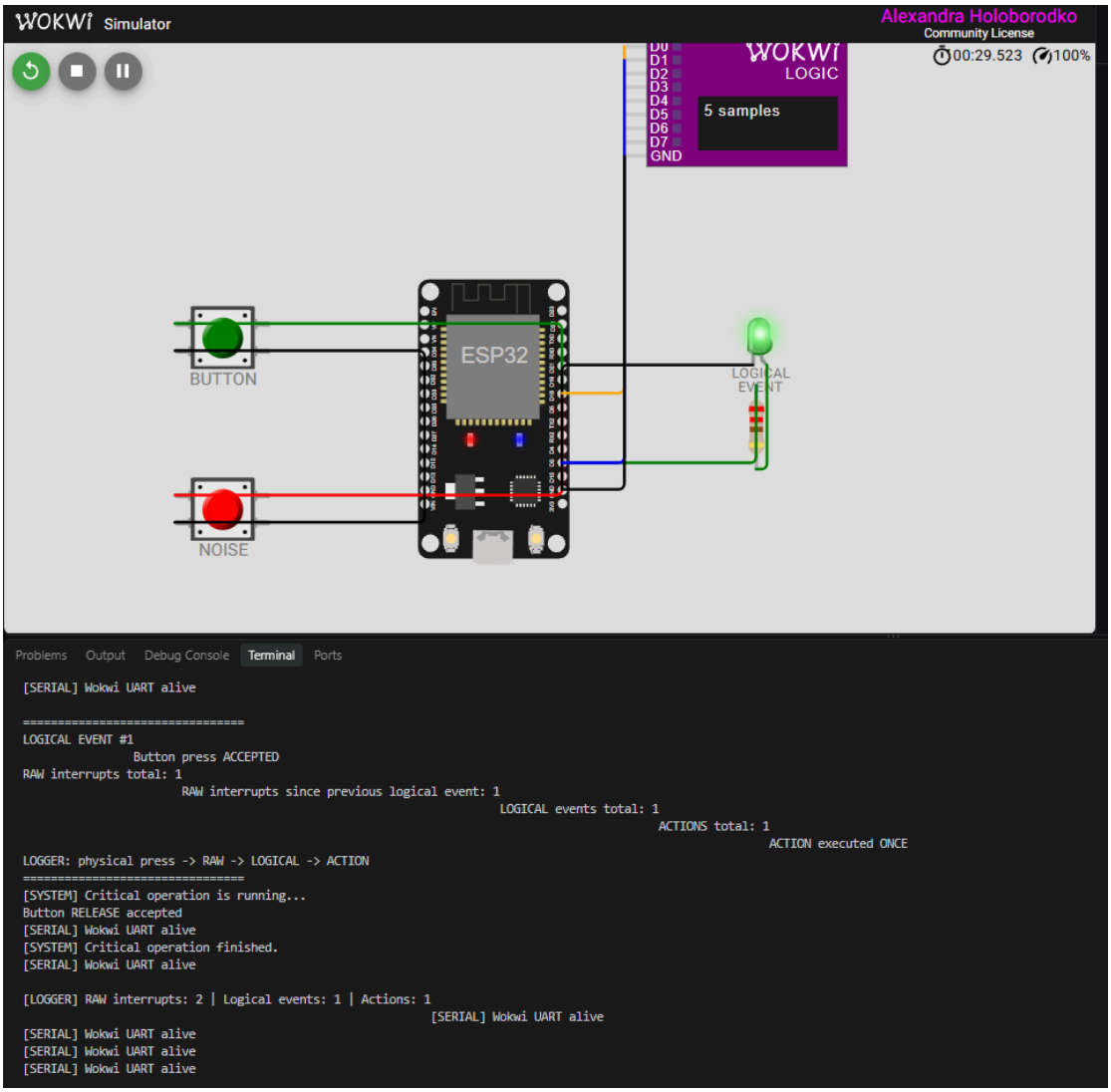


Figure: TC-05 Short noise pulse

Bug_2 (TC-06) — Extra logical event/action is present after pressing the NOISE button during a critical operation

Environment	ESP32 DevKit v1
Preconditions	Simulation is started
Steps to Reproduce	1. Press the "Button" button. 2. Release the "Button" button. 3. Press the "NOISE" button during the critical operation. 4. Release the "NOISE" button. 5. Check the Serial Monitor.
Expected	No false logical press/action
Actual	A logical press/action is present after pressing the NOISE button
Evidence	TC-06 Noise during critical operation (screenshot below)
Reproducibility	Always
Severity	Critical

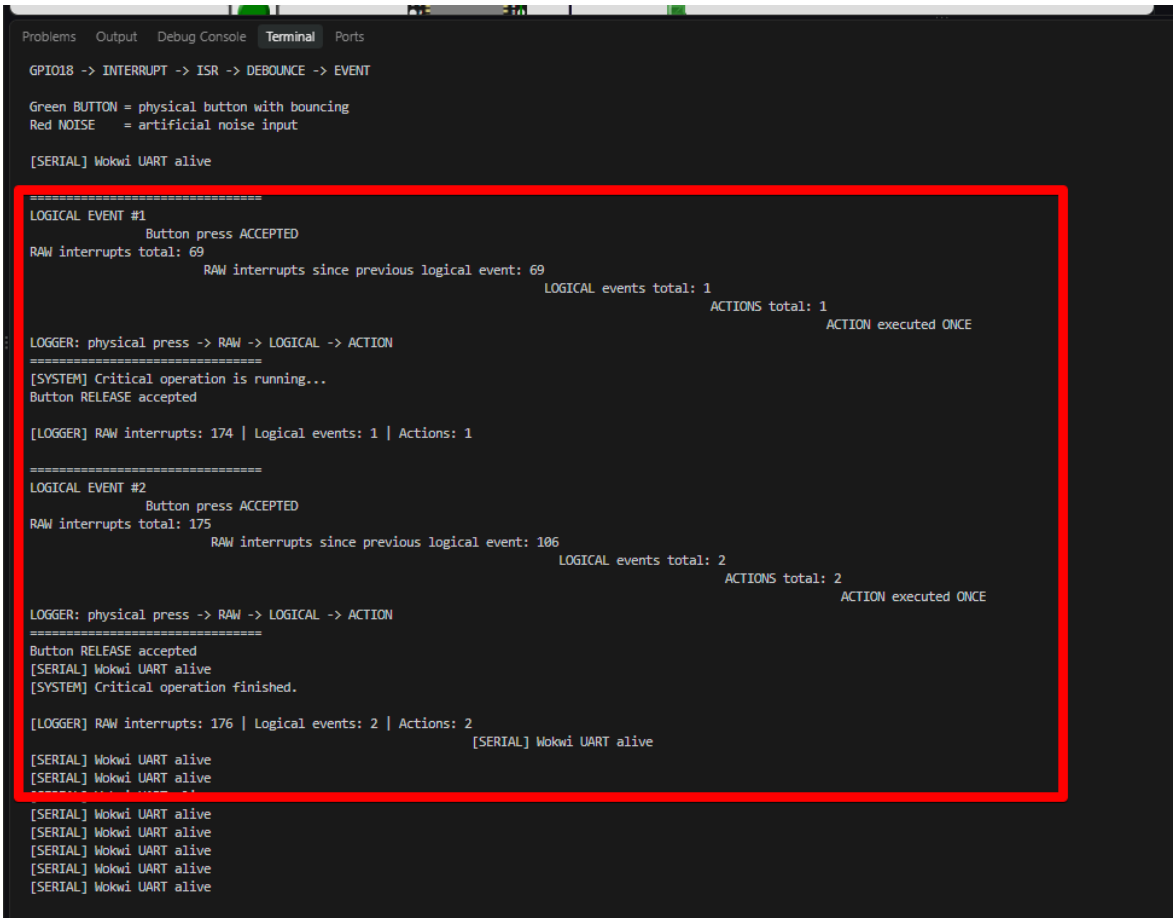


Figure: TC-06 Noise during critical operation